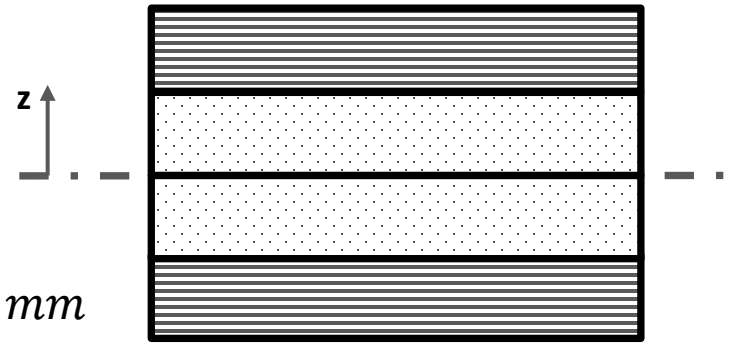
Structures and Materials

Tutorial | Dr.-Ing. Andreas Hermanutz

Composites

1. Composites - CLT



Task 1: Determine the constitutive law for a laminate with a layup of $[0\ /\ 90]_S$

$E_1 = 132680\ MPa$ $E_2 = 9370$ MPa $G_{12} = 9642$ MPa $\nu_{12} = 0.287$ $t = 0.2\ mm$

Step 1: Determining the stiffness matrix Q for a 0° laminate.

$$[Q] = \begin{bmatrix} \frac{E_1}{1-\nu_{12}\nu_{21}} & \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}} & 0 \\ \frac{\nu_{21}E_1}{1-\nu_{12}\nu_{21}} & \frac{E_2}{1-\nu_{12}\nu_{21}} & 0 \\ 0 & 0 & G_{12} \end{bmatrix} = \begin{bmatrix} 133456 & 2704 & 0 \\ 2704 & 9424 & 0 \\ 0 & 0 & 9642 \end{bmatrix}$$

$$\nu_{21} = \nu_{12} \cdot \frac{E_2}{E_1}$$

1. Composites - CLT

Step 2: Transfer the 0° stiffness matrix to 90° stiffness matrix.

$$[\mathrm{T}] = \begin{bmatrix} cos^2(90^0) & sin^2(90^0) & 2\sin(90^0)\cos(90^0) \\ sin^2(90^0) & cos^2(90^0) & -2\sin(90^0)\cos(90^0) \\ -\sin(90^0)\cos(90^0) & \sin(90^0)\cos(90^0) & cos^2(90^0) - sin^2(90^0) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$[T]^{-1} = \begin{bmatrix} cos^2(90^0) & sin^2(90^0) & -2\sin(90^0)\cos(90^0) \\ sin^2(90^0) & cos^2(90^0) & 2\sin(90^0)\cos(90^0) \\ \sin(90^0)\cos(90^0) & -\sin(90^0)\cos(90^0) & cos^2(90^0) - sin^2(90^0) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

1. Composites - CLT

Step 2: Transfer the 0° stiffness matrix to 90° stiffness matrix.

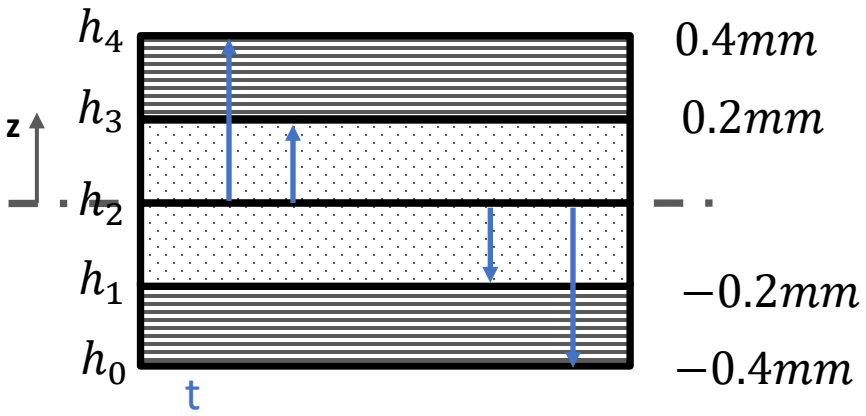
$$[\bar{Q}]_{90} = [T]^{-1}[Q][T] = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 133456 & 2704 & 0 \\ 2704 & 9424 & 0 \\ 0 & 0 & 9642 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$[\bar{Q}]_{90} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2704 & 133456 & 0 \\ 9424 & 2704 & 0 \\ 0 & 0 & -9642 \end{bmatrix} = \begin{bmatrix} 9424 & 2704 & 0 \\ 2704 & 133456 & 0 \\ 0 & 0 & 9642 \end{bmatrix}$$

1. Composites - CLT

Step 3: Assembly ABD matrix: A Matrix

$$A_{ij} = \sum_{k=1}^{n} [\bar{Q}_{ij}]_k (h_k - h_{k-1})$$

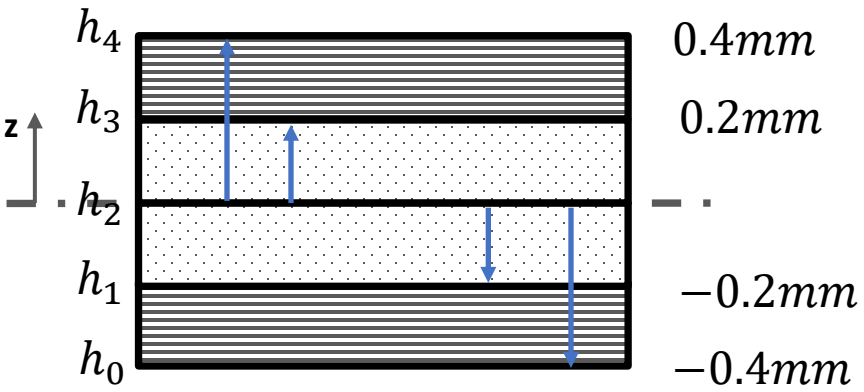


$$A = \begin{bmatrix} 133456 & 2704 & 0 \\ 2704 & 9424 & 0 \\ 0 & 0 & 9642 \end{bmatrix} \overbrace{(-0.2-(-0.4))}^{t} + \begin{bmatrix} 9424 & 2704 & 0 \\ 2704 & 133456 & 0 \\ 0 & 0 & 9642 \end{bmatrix} \overbrace{(0.0-(-0.2))}^{t}$$
$$+ \begin{bmatrix} 9424 & 2704 & 0 \\ 2704 & 133456 & 0 \\ 0 & 0 & 9642 \end{bmatrix} \overbrace{(0.2-0.0)}^{t} + \begin{bmatrix} 133456 & 2704 & 0 \\ 2704 & 9424 & 0 \\ 0 & 0 & 9642 \end{bmatrix} \overbrace{(0.4-0.2)}^{t}$$
$$= \begin{bmatrix} 57152 & 2163 & 0 \\ 2163 & 57152 & 0 \\ 0 & 0 & 7713 \end{bmatrix}$$

1. Composites - CLT

Step 3: Assembly ABD matrix: B Matrix

$$B_{ij} = \frac{1}{2}\sum_{k=1}^{n}\left[\bar{Q}_{ij}\right]_k\left(h_k^2 - h_{k-1}^2\right)$$



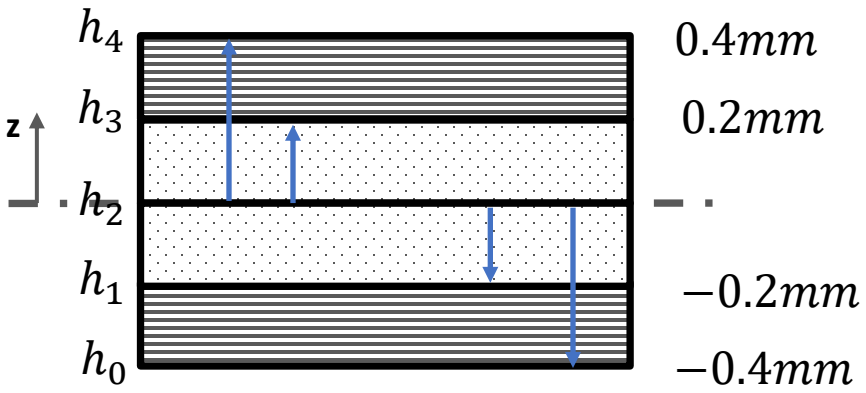
$$B = \frac{1}{2}\begin{bmatrix}133456 & 2704 & 0\\ 2704 & 9424 & 0\\ 0 & 0 & 9642\end{bmatrix}\underbrace{-\left((-0.4)^2-(-0.2)^2\right)}_{\left((-0.2)^2-(-0.4)^2\right)} + \frac{1}{2}\begin{bmatrix}9424 & 2704 & 0\\ 2704 & 133456 & 0\\ 0 & 0 & 9642\end{bmatrix}\left(0.0-(-0.2)^2\right)$$

$$+\frac{1}{2}\begin{bmatrix}9424 & 2704 & 0\\ 2704 & 133456 & 0\\ 0 & 0 & 9642\end{bmatrix}\left((0.2)^2-0.0\right) + \frac{1}{2}\begin{bmatrix}133456 & 2704 & 0\\ 2704 & 9424 & 0\\ 0 & 0 & 9642\end{bmatrix}\left((0.4)^2-(0.2)^2\right) = [0]$$

1. Composites - CLT

Step 3: Assembly ABD matrix: B Matrix

$$D_{ij} = \frac{1}{3}\sum_{k=1}^{n}\left[\bar{Q}_{ij}\right]_k\left(h_k^3 - h_{k-1}^3\right)$$



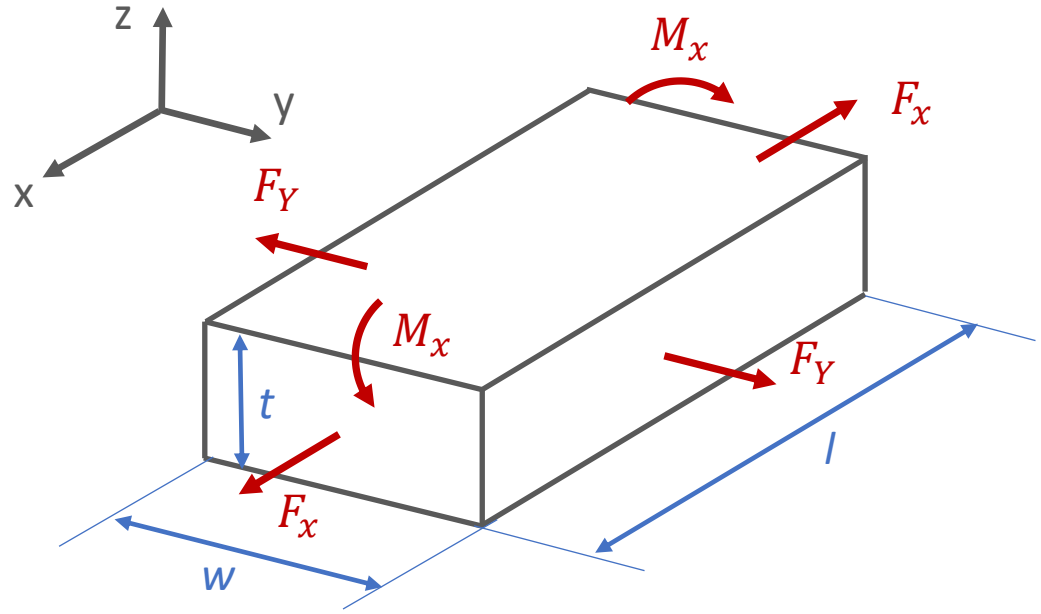
$$D = \frac{1}{3}\begin{bmatrix}133456 & 2704 & 0\\ 2704 & 9424 & 0\\ 0 & 0 & 9642\end{bmatrix}\left((-0.2)^3-(-0.4)^3\right) + \frac{1}{3}\begin{bmatrix}9424 & 2704 & 0\\ 2704 & 133456 & 0\\ 0 & 0 & 9642\end{bmatrix}\left(0.0-(-0.2)^3\right)$$

$$+\frac{1}{3}\begin{bmatrix}9424 & 2704 & 0\\ 2704 & 133456 & 0\\ 0 & 0 & 9642\end{bmatrix}\left((0.2)^3-0.0\right) + \frac{1}{3}\begin{bmatrix}133456 & 2704 & 0\\ 2704 & 9424 & 0\\ 0 & 0 & 9642\end{bmatrix}\left((0.4)^3-(0.2)^3\right)$$

$$D = \begin{bmatrix}5032 & 115 & 0\\ 115 & 1063 & 0\\ 0 & 0 & 411\end{bmatrix}$$

1. Composites - CLT

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \begin{bmatrix} \overset{[N\,/\,mm]}{\begin{matrix} 57152 & 2163 & 0 \\ 2163 & 57152 & 0 \\ 0 & 0 & 7713 \end{matrix}} & \overset{[N]}{\begin{matrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{matrix}} \\ \begin{matrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{matrix} & \underset{[N\,mm]}{\begin{matrix} 5032 & 115 & 0 \\ 115 & 1063 & 0 \\ 0 & 0 & 411 \end{matrix}} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \\ \kappa_x \\ \kappa_y \\ \kappa_{yx} \end{Bmatrix}$$



$$N_x = \frac{F_x}{w} \quad [N\,/\,mm]$$

$$N_y = \frac{F_y}{L} \quad [N\,/\,mm]$$

$$M_x = \frac{M_x}{w} \quad [Nmm\,]$$

Physical interpretation of D11 and D22

$D_{11} > D_{22}$

0° have larger distance from mi-plane, than 90° layers → higher second moment of area → higher impact on bending stiffness

2. eLamX Exercise

The use of eLamX is not relevant for the exam, but the calculation will help to strengthen your understanding.

Task 1: Create the UD Material Model

Task 2: Create a laminate with 1 ply:
Check the ABD matrix
Check the stiffness polar plot

Task 3: Set-up the laminate of Chapter 1 and double confirm the results

2. eLamX Exercise

Special features of the ABD matrices: All Laminates are symmetric → $[B] = 0$ no bending-tension coupling

General: $[\pm 20 \,/\, 60]_S$

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{21} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{21} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix}$$

Balanced: $[\pm 15 / \pm 30]_S$

Angle-Ply: $[\pm 20]_S$

$$A = \begin{bmatrix} A_{11} & A_{12} & 0 \\ A_{21} & A_{22} & 0 \\ 0 & 0 & A_{66} \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{21} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix}$$

no shear-tension coupling

Cross-Ply: $[0 \,/\, 90 \,/\, 0]_S$

$$A = \begin{bmatrix} A_{11} & A_{12} & 0 \\ A_{21} & A_{22} & 0 \\ 0 & 0 & A_{66} \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} D_{11} & D_{12} & 0 \\ D_{21} & D_{22} & 0 \\ 0 & 0 & D_{66} \end{bmatrix}$$

no bend-twist coupling

2. eLamX Exercise

Special features of the ABD matrices: All Laminates are symmetric → $[B] = 0$

Special features of the

$$[\pm 30/\pm 60/\ 0\ /\ 90]_S$$

$$A = \begin{bmatrix} A_{11} & A_{12} & 0 \\ A_{21} & A_{22} & 0 \\ 0 & 0 & A_{66} \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad D = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{21} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix}$$

→ Compare with Aluminium Plate

2. Type of Laminates Summary

Based on the stacking sequence, composite laminates can be categorized as:

Angle-ply Laminate
A laminate is called an angle-ply laminate if it has layers of the same thickness and material, and are oriented at θ and −θ.

Cross-ply Laminate
A laminate is called a cross-ply laminate if all the layers have the orientation 0° and 90°.

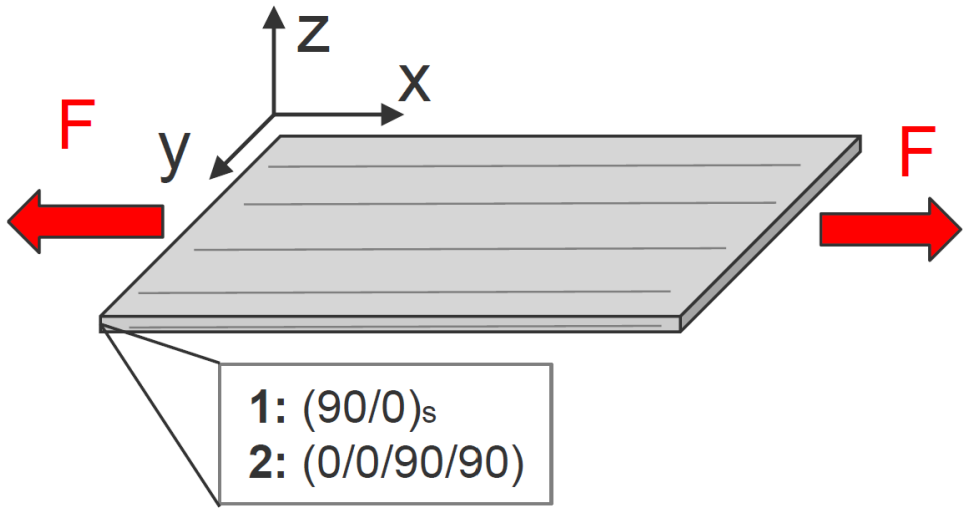
Balanced Laminate
A laminate is called a balanced laminate when it has pairs of layers of the same thickness and material, and the angles of the layers have opposite signs. Balanced laminates can also have layers at 0° and 90°.

Symmetric Laminate
A laminate is called symmetric when the material, angle, and thickness of the layers are the same above and below the midplane.

Antisymmetric Laminate
A laminate is called antisymmetric when the material and thickness of the layers are the same above and below the midplane, but the orientation of the layers have opposite signs above and below the midplane.

3. eLamX Exercise – Plate under tension

A suitable laminate is to be chosen for the tensile loaded plate shown on the right side. The material is C(HM)-EP. The maximum allowed plate thickness is $t = 1.0mm$.



Task 1: Which stiffness is relevant for this load type?

The membrane stiffness must be used for further calculation because the load effects in the laminate plane.

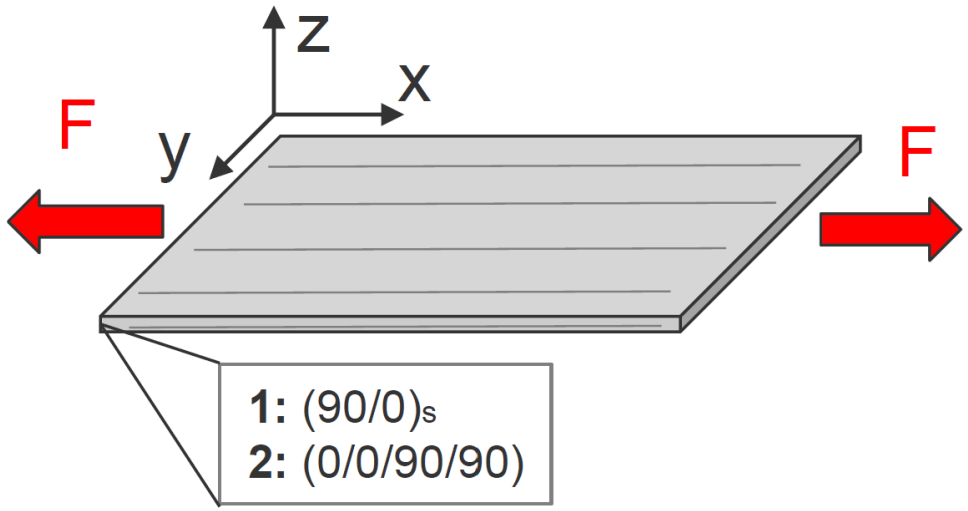
Task 2: Which laminate will presumably exhibit a higher stiffness E_x in load direction?

In both laminates the same single layers are applied (2x0°-layer, 2x90°-layer). Therefore, it is presumed that the membrane stiffness in both laminates will be equivalent.

Task 3: Calculate the relevant stiffnesses (engineering constants) for both laminates and compare your estimations from with the calculated values.

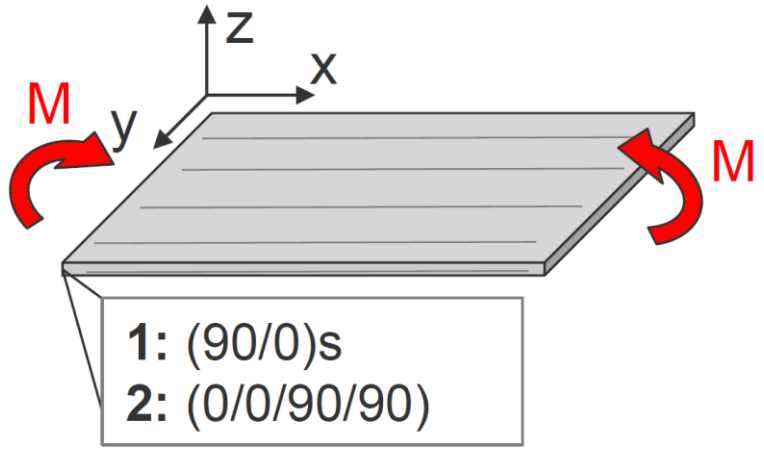
3. eLamX Exercise – Plate under tension

Task 3: Calculate the relevant stiffnesses (engineering constants) for both laminates and compare your estimations from with the calculated values.



	with Transverse contraction restriction	without Transverse contraction restriction
$[90\ /\ 0]_S$	118512 MPa	118489 MPa
[0 / 0 / 90 / 90]	118512 MPa	39246 MPa

3. eLamX Exercise – Plate under tension



The same plate as in the previous task is loaded by a bending moment instead of the tensile force. The laminate and the material are the same.

Task 1: Would you use the membrane stiffness for analysis again in this case?

No, in this case the bending stiffness matrix coefficient D11 of the plate is relevant.

Task 2: Will the relevant stiffnesses be independent from the lay-up?

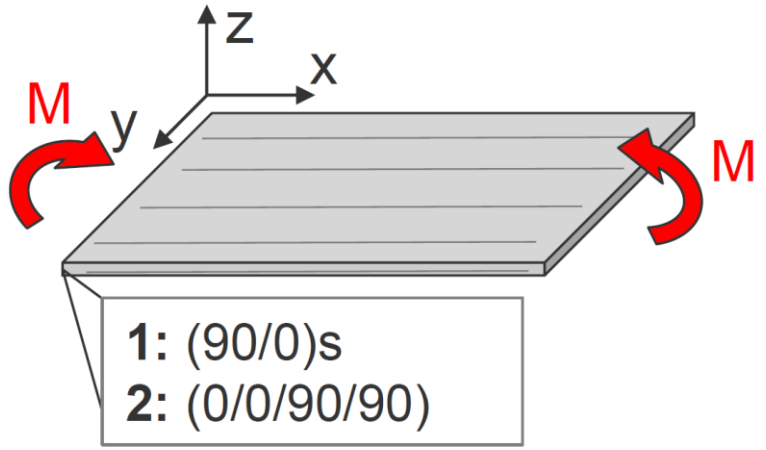
For laminates which are loaded with bending, the lay-up plays an important role. Layers further away from the symmetry plane have a higher influence on bending resistance.

Task 3: Calculate the relevant values for the stiffness for both laminates and compare the stiffnesses with each other.

Task 4: Which lay-up could improve the relevant stiffness for the given load?

3. eLamX Exercise – Plate under tension

The same plate as in the previous task is loaded by a bending moment instead of the tensile force. The laminate and the material are the same.



Task 3: Calculate the relevant values for the stiffness for both laminates and compare the stiffnesses with each other.

	$[90\ /\ 0]_S$	$[0\ /\ 0\ /\ 90\ /\ 90]$
D11 [Nmm]	2882	9876

Task 4: Which lay-up could improve the relevant stiffness for the given load?

A significant improvement of the bending stiffness can be achieved by shifting the 0°-layers of the symmetry plane to the outer layers. Outer layers have a higher influence on the second moment of area (higher distance from mid plane).

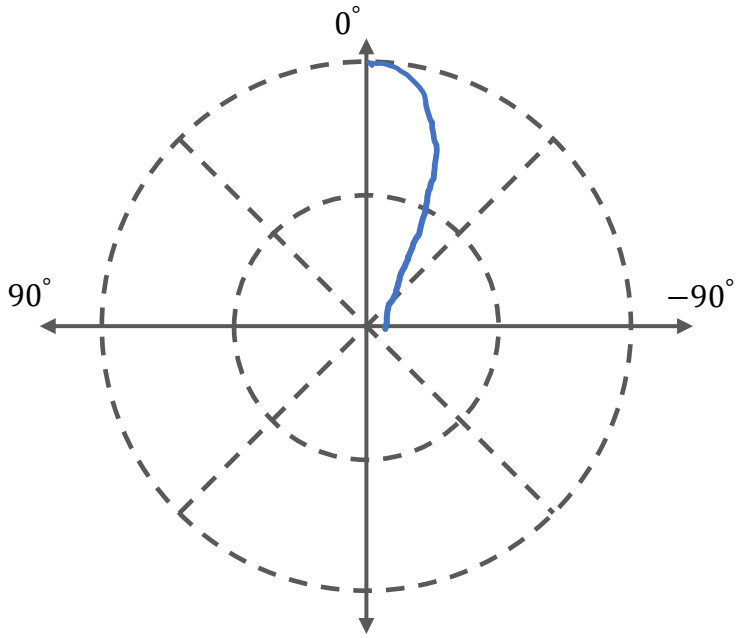
	$[0\ /\ 90]_S$
D11 [Nmm]	16869

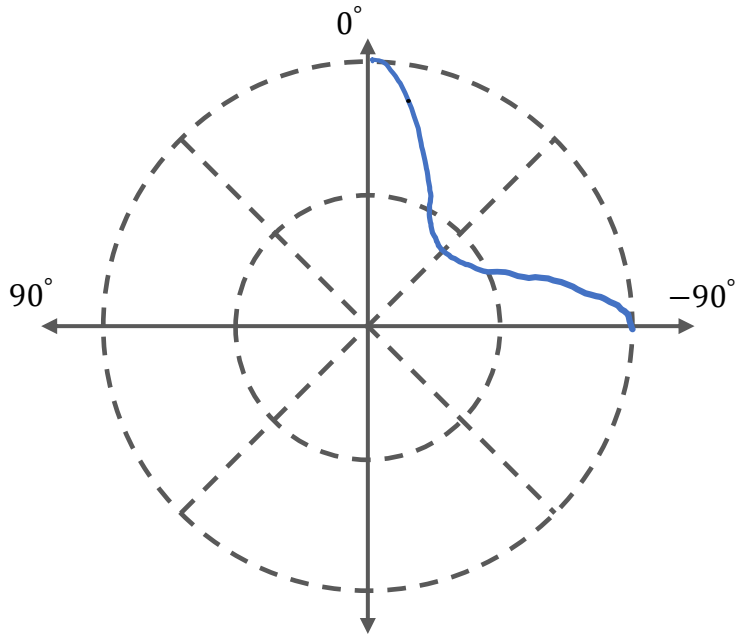
4. eLamX Exercise – Polar Diagram

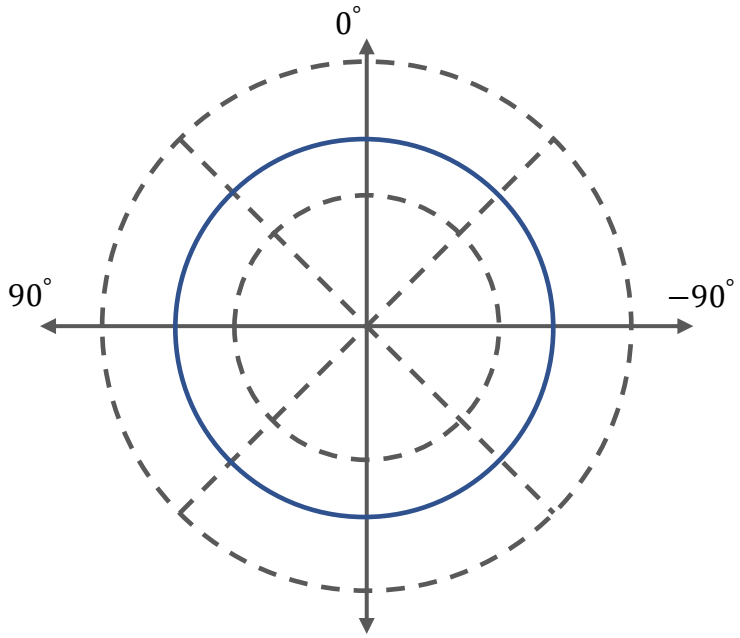
Three different laminates each with C(HM)-EP material with ply thickness 0.125mm are given:

1. UD-Laminate
2. $[0\ /\ 90]_S$
3. $[0\ /\ 45\ /\ -45\ /\ 90\ ]_S$

Task 1: Sketch the curve of the stiffnesses E_x in the below polar diagrams for laminate 1 – 3 and confirm it with the results from eLamX.





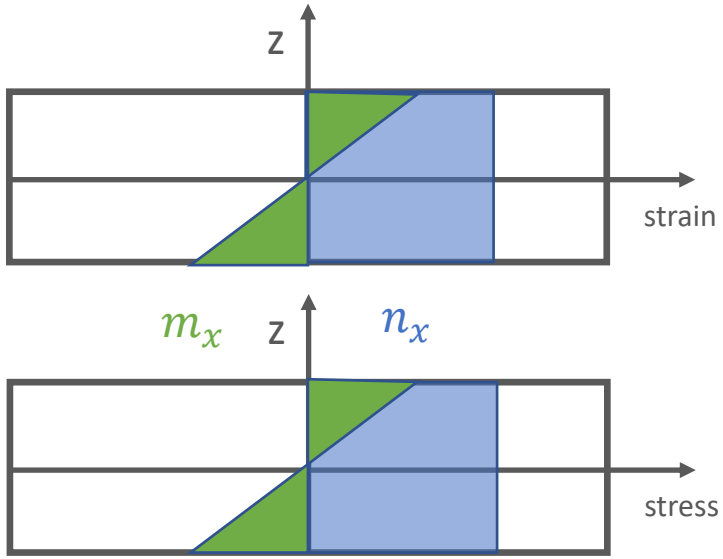


4. eLamX Exercise – Strain and Stress Distribution

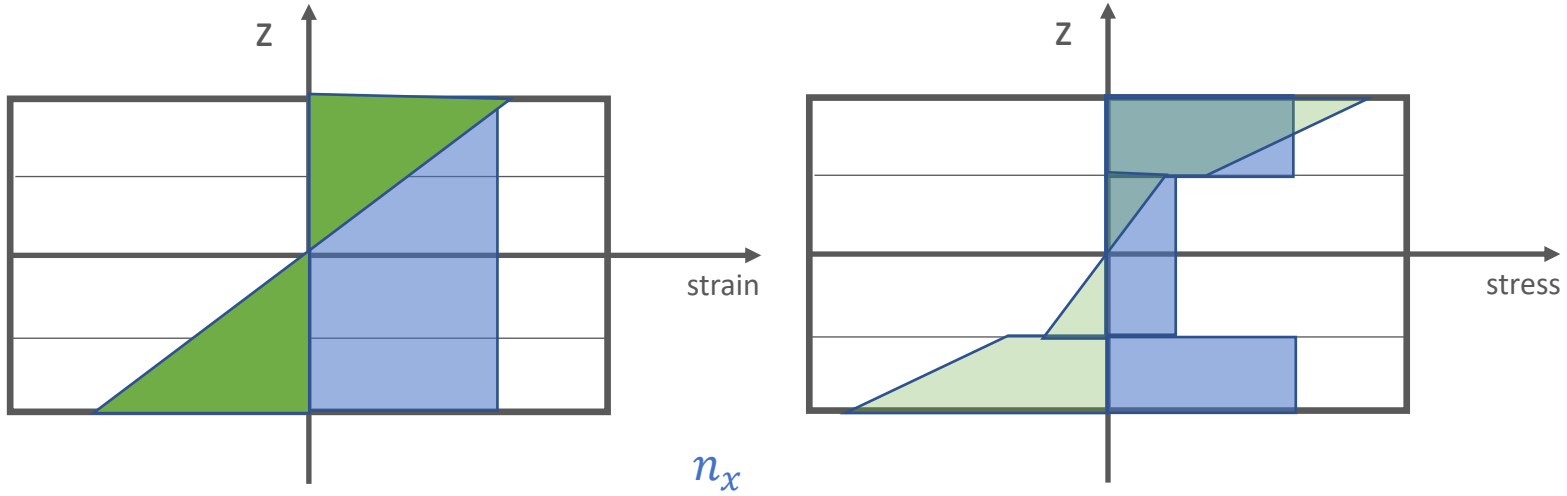
Two laminates each with C(HM)-EP material with ply thickness 0.125mm are given:

1. UD-Laminate
2. $[0\ /\ 90]_S$

Task 1: Sketch the in-plane stress under a line load of $n_x = 1\ N/mm$ and $m_x = 1\ Nmm$.



UD-ply



Laminate

5. Aluminium vs. Black Metal Design Composite Plate using CLT

Compare the structural performance of an aluminium plate with a quasi-isotropic Black Metal Design composite laminate using a Classical Laminate Theory (CLT).

Geometry: Square Plate 300mm x 300mm x 5mm; CFRP: C-AS-4 | EP-3501-6, Aluminium 7075-T6: $\sigma_{max} = 500MPa, E = 72GPa, \nu = 0.33$

Task 1 – Aluminium Reference

- Calculate the tensile/bending stiffness.
- Select an appropriate failure criterion.
- Determine the failure load.
- Calculate the plate mass.

Task 2 – Composite Plate

Design a **symmetric quasi-isotropic laminate**.

- Calculate the laminate tensile/bending stiffness.
- Choose an appropriate composite failure criterion (e.g. Tsai-Wu or Maximum Stress).
- Determine the laminate failure load.
- Calculate the laminate mass.

Task 3 – Discussion

Complete the following comparison table.

- Which structure is stiffer?
- Which structure carries the highest load?
- Which structure is lighter?
- Which design provides the highest structural efficiency?
- Was the chosen failure criterion appropriate?

	Aluminium	CFRP
Failure criterion		
Plate Tensile Stiffness		
Plate Bending Stiffness		
Failure load		
Plate mass		
Specific Plate Tensile Stiffness		
Specific Plate Bending Stiffness		
Specific strength		